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Fatigue properties of Fe-30.5Mn-8Al-1C austenitic low-density steel: Critical impact of κ-carbide precipitation state

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ABSTRACT

This study systematically investigates the tension–tension fatigue behavior and deformation mechanisms of solution-treated and aging-treated Fe-30.5Mn-8Al-1C (wt%) austenitic low-density steels, focusing on the critical role of κ-carbide precipitation state in controlling fatigue properties. In aged samples, intragranular κ-carbides induce planar dislocation slip through a "glide plane softening" mechanism, enhancing slip reversibility under cyclic loading and thereby improving fatigue life. Strengthening is primarily due to the interaction between dislocations and intragranular κ-carbides. An appropriate increase in the size of intragranular κ-carbides significantly enhances fatigue life and fatigue strength at low stress amplitudes. Conversely, intergranular κ-carbide precipitation impedes slip transmission, intensifies localized stress concentration, and accelerates damage, thus reducing fatigue life at high stress amplitudes. These findings strongly demonstrate that accelerating the precipitation of intragranular κ-carbides while suppressing intergranular precipitation is an effective microstructural pathway to concurrently enhance fatigue performance of Fe-Mn-Al-C austenitic low-density steels across the entire range of stress amplitudes.

1. Introduction

Driven by increasingly stringent emission regulations and fuel efficiency demands, the pursuit of automotive lightweighting has intensified the search for materials with superior strength-to-weight ratios. Low-density steels, especially Fe-Mn-Al-C alloys, have emerged as promising candidates [1–5]. The addition of ~10 wt% aluminum reduces density by approximately 10–13 % compared with conventional steels, while maintaining or even enhancing strength–ductility synergy [3,6–8]. Some previous studies have shown that optimized aging treatment facilitates the uniform precipitation of high-density nanoscale $(Fe,Mn)_3AlC$ κ-carbides within the γ matrix through spinodal decomposition [9,10]. These intragranular κ-carbides provide pronounced precipitation hardening [11–15]. However, high-temperature exposure also triggers discontinuous precipitation and cellular transformation at grain boundaries (GBs), leading to the formation of coarse intergranular κ-carbides [16,17]. Such precipitates would concentrate interfacial stress, degrade mechanical properties, and significantly compromise alloy reliability [2,18]. Accordingly, tailoring κ-carbide precipitation to achieve intragranular refinement while suppressing intergranular precipitate coarsening is of critical importance for the performance optimization.

Fatigue failure constitutes an important failure mode in automotive structures and components [19,20]. As vehicle weight is reduced, ensuring adequate fatigue life and maintaining high strength and ductility becomes equally critical for safety and reliability. Consequently, understanding the fatigue performance of advanced low-density steels is of considerable engineering significance. Gomez et al. [21] reported that intragranular κ-carbides provide precipitation strengthening through dislocation shearing, significantly improving the high-cycle fatigue (HCF) performance in Fe-29Mn-8.7Al-1C. Wegener et al. [22] conducted a low-cycle fatigue (LCF) study on Fe-29.8Mn-7.65Al-1.11C steel, and found that intragranular κ-carbides improved fatigue life at a low strain amplitude but reduced fatigue life at a high strain amplitude due to aggravated cyclic softening. Li et al. [23] performed tension–compression fatigue tests on solution-treated Fe-26Mn-10Al-C alloys, observing rapid initial cyclic softening, followed by cyclic hardening and eventual performance degradation. They attributed the hardening to the dynamic precipitation of κ-carbides during cyclic deformation. Currently, investigations on the fatigue behavior of Fe-Mn-Al-C austenitic steels still remain relatively limited, primarily focusing on the influence of intragranular κ-carbides. However, the mechanisms

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by which different intragranular κ-carbide precipitation states affect fatigue performance are still poorly understood, particularly regarding the role of intergranular κ-carbides.

The present study systematically investigates the fatigue behavior of an Fe-30.5Mn-8Al-1C (wt%) alloy under tension–tension cyclic loading. By tailoring the morphology and distribution of κ-carbide precipitation through controlled thermal treatments, the effects of both intragranular and intergranular κ-carbides on the fatigue life, cyclic response, and damage mechanisms are elucidated. This work is expected to provide fundamental insights crucial for the development of fatigue-resistant, high-strength, and low-density steels that are suitable for automotive structural applications under severe cyclic loading conditions.

2. Experimental materials and methods

2.1. Materials

The investigated steel has a nominal composition of Fe-30.5Mn-8Al-1C (wt%). A 50 kg ingot was prepared by vacuum melting and subsequently forged into a billet with a cross-section of 30 mm × 30 mm. The billet was then hot-rolled to a final thickness of 4 mm. Solution treatment was conducted at 1050 °C for 1 h, followed by water quenching to room temperature; this condition is referred to as ST. To obtain different precipitation states of intragranular and intergranular κ-carbides, selected ST samples were further aged at 550 °C, 600 °C, and 650 °C for 6 h, designated as 550-6, 600-6, and 650-6, respectively.

2.2. Mechanical tests

Flat dog-bone-shaped tensile samples were machined from the sheets using wire electrical discharge machining (EDM), with their tensile axes parallel to the rolling direction. A schematic of the shape and dimensions of the tensile test samples is provided in Fig. 1. The sample surfaces were mechanically polished and subsequently electro-polished in a solution of 8 vol.% perchloric acid in ethanol for 30 s at 25 V to remove surface stresses and scratches. Uniaxially tensile tests were performed at room temperature using a universal testing machine (Shimadzu AG-Xplus 250 kN) under a constant strain rate of 10^{-3} s^{-1}.

Flat dog-bone-shaped fatigue samples were also extracted from the sheets via EDM, with the loading axis parallel to the rolling direction. These samples shared the same dimensions as the tensile samples (Fig. 1). The surfaces were prepared using the same procedure of mechanical grinding and electropolishing to ensure a stress-free and smooth finish. Stress-controlled tension–tension fatigue tests were conducted in ambient air using a servo-hydraulic fatigue testing machine (Shimadzu EHF-EV050K2A-020-1). A sinusoidal waveform was adopted in fatigue tests at a frequency of 20 Hz and a stress ratio of $R = 0.1$.

2.3. Microstructural characterizations

The microstructure of the undeformed ST sample was characterized by electron backscatter diffraction (EBSD) on a TESCAN MIRA3 scanning electron microscope (SEM) equipped with an Oxford EBSD detector. Prior to observations, the sample surface was prepared through mechanical polishing followed by electropolishing. Transmission electron microscopy (TEM) was conducted on a JEOL JEM-2100F microscope operating at 200 kV to characterize the precipitation of κ-carbides in the aged samples (550-6, 600-6 and 650-6). TEM foils were prepared by mechanically grinding slices to approximately 50 μm in thickness, followed by twin-jet electropolishing in a solution of 8 vol.% perchloric acid in ethanol at 30 V. The electrolyte was maintained at approximately −20 °C to prevent sample oxidation.

For the post-fractured samples, the fracture surfaces and lateral surfaces were examined using a JEOL JSM-7800F SEM. Furthermore, EBSD was employed to analyze crack propagation path on external surfaces. These scans were acquired using a step size of 0.8 μm, and the data were analyzed using AZtecCrystal post-processing software. To gain further insight into the deformation mechanisms, the deformation microstructures in the post-fractured samples were characterized by TEM. TEM foils were extracted from the gauge sections of the deformed specimens to ensure microstructural representativeness.

3. Results

3.1. Microstructures prior to deformation

Fig. 2 shows the microstructures in the ST and aged samples. The inverse pole figure (IPF) map of the ST sample reveals a fully austenitic microstructure (Fig. 2a). After aging, the 550-6 sample exhibits a high density of nanoscale κ-carbides (~11 nm) with rectangular morphology, uniformly precipitated within the austenite grains (Fig. 2b). With increasing aging temperature, the intragranular κ-carbides in the 600-6 sample coarsen to ~18 nm, and evolve into larger rectangular parallelepipeds to form well-defined arrays aligned along the orthogonal <100> directions of the matrix (Fig. 2c). Concurrently, intergranular κ-carbide precipitation is also observable. Further increasing aging temperature to 650 °C leads to coarsening of intragranular κ-carbides to approximately 24 nm, accompanied by more pronounced intergranular precipitation in the 650-6 sample (Fig. 2d). The size of intragranular κ-carbides is quantified using an equivalent-circle diameter, and the quantitative evolution of κ-carbide size with aging temperature is summarized in Fig. 3. The detailed κ-carbide precipitation behavior of the Fe-30.5Mn-8Al-1C alloy during aging treatment has been reported in our previous work [6]. This microstructural overview provides the basis for interpreting the subsequent fatigue behavior.

3.2. Tensile properties

Engineering stress–strain curves for the ST and aged samples are shown in Fig. 4, and the corresponding data of tensile properties are listed in Table 1. All aged conditions demonstrate significant increases in both yield and ultimate tensile strengths but a reduction in ductility,

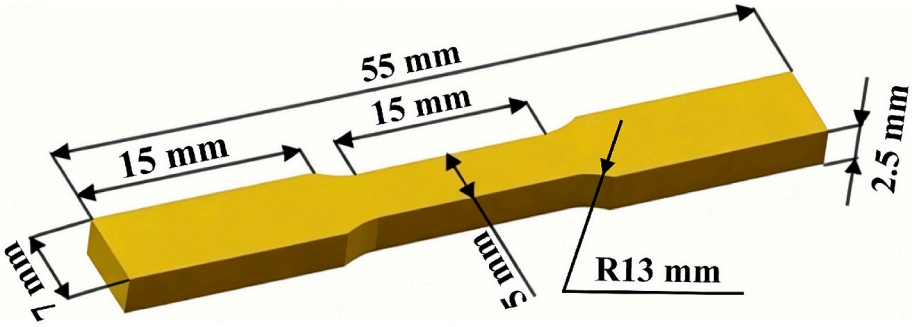

Fig. 1. Schematic of sample shape and dimensions for tensile and fatigue tests.

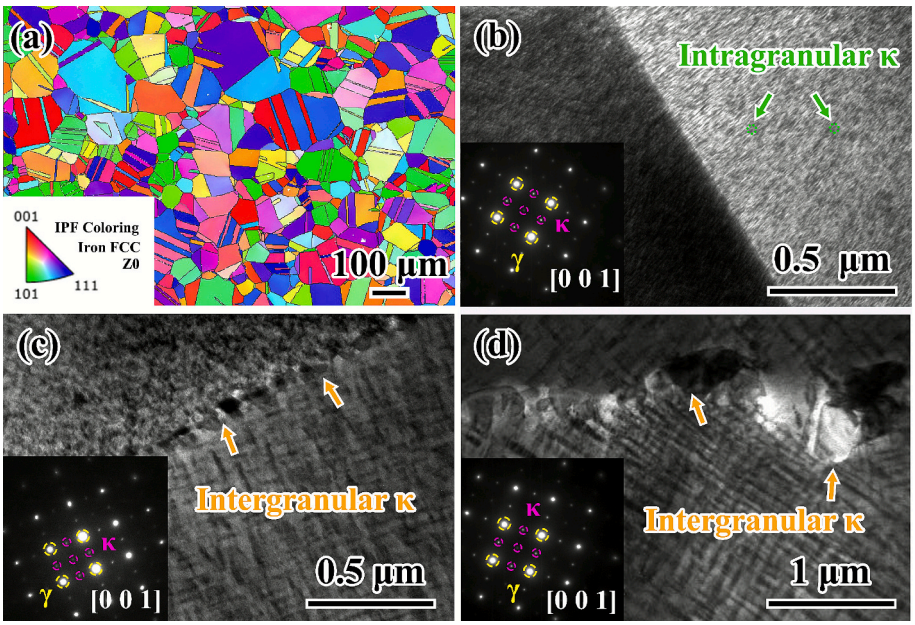

Fig. 2. IPF map of the ST sample (a), and bright-field TEM micrographs showing the microstructures in the aged samples of 550-6 (b), 600-6 (c), and 650-6 (d).

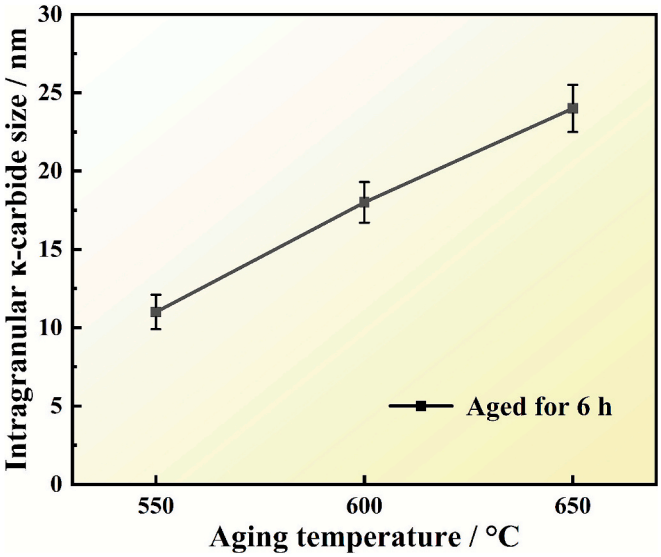

Fig. 3. Size of intragranular κ-carbide precipitates as a function of aging temperature after 6 h of aging. Data are presented as mean ± standard deviation.

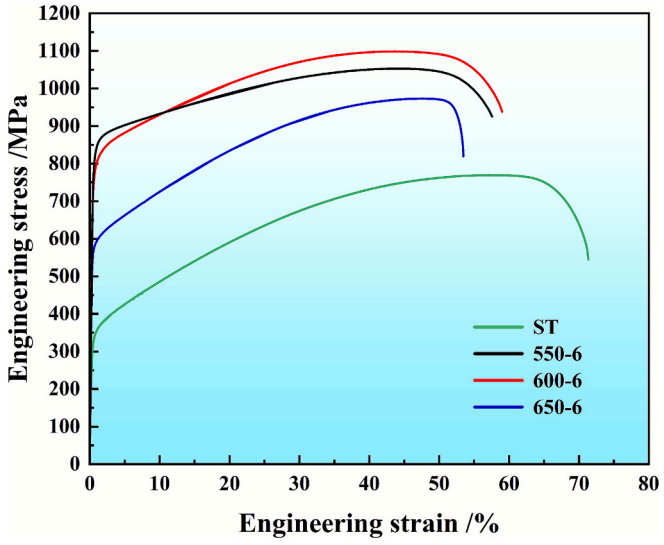

Fig. 4. Tensile engineering stress–strain curves of the ST, 550-6, 600-6, and 650-6 samples.

compared with the ST condition. Among the aged samples, 550-6 and 600-6 samples demonstrate superior strengths, whereas the 650-6 sample shows a marked reduction in strength. The strengthening mechanism primarily arises from precipitation hardening by intragranular κ-carbides through their interactions with dislocations during deformation. Conversely, the diminished strength in the 650-6 sample is attributed to grain boundary weakening caused by serious intergranular κ-carbide precipitation [6,24].

3.3. Tension-tension fatigue properties

For the present stress-controlled tension–tension fatigue tests, the finite-life regime was analyzed using the Basquin equation [25]:

$$\Delta\sigma/2 = \sigma'_f(2N_f)^b \quad (1)$$

where N_f is the fatigue life, $\Delta\sigma/2$ is the stress amplitude, σ'_f is the fatigue strength coefficient reflecting the fatigue resistance at a specific stress level, and b is the fatigue strength exponent characterizing the sensitivity to cumulative fatigue damage [26].

According to Eq. (1), the $\Delta\sigma/2$-$2N_f$ curves of the Fe-30.5Mn-8Al-1C steel are approximately linear in the finite-life region under double-logarithmic coordinates, as shown in Fig. 5. The fitted fatigue strength coefficient σ'_f and exponent b, together with the root mean square error of $\log_{10} N_f$, are summarized in Table 1. At a given stress amplitude, all aged samples exhibit longer fatigue lives compared with the ST sample. The fatigue strength, defined as the maximum stress amplitude sustained for 10^7 cycles without failure, was measured to be 157.5 MPa for ST, 162 MPa for 550-6, 175.5 MPa for 600-6, and 171 MPa for 650-6. Apparently, the aging treatment effectively enhances the fatigue resistance at low stress amplitudes, with the 600-6 sample delivering the highest fatigue strength. It is further noted that the 550-6 condition exhibits the longest fatigue lives among the aged samples at higher stress amplitudes, whereas at lower stress amplitudes close to the fatigue limit,

Table 1
Tensile and fatigue properties of the ST, 550-6, 600-6 and 650-6 samples.

Samples	YS	UTS	UE	TE	σ'_f	b	RMSE
ST	337 ± 32	784 ± 15	57.7 ± 1.3	72.0 ± 0.8	1510	−0.15	0.0429
550-6	829 ± 17	1045 ± 8	45.5 ± 1.4	57.7 ± 0.4	2243	−0.17	0.0952
600-6	776 ± 19	1104 ± 17	45.1 ± 1.5	58.4 ± 1.4	1040	−0.11	0.1776
650-6	579 ± 14	978 ± 4	46.4 ± 1.2	52.7 ± 1.6	1153	−0.12	0.1139

YS: yield strength (MPa); UTS: ultimate tensile strength (MPa); UE: uniform elongation (%); TE: total elongation (%); σ'_f: fatigue strength coefficient; b: fatigue strength exponent; RMSE: root mean square error of $\log_{10} N_f$.

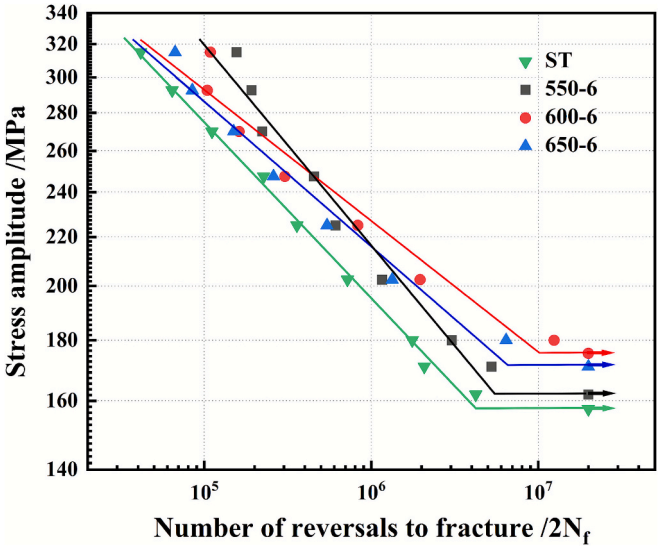

Fig. 5. $\Delta\sigma/2$-$2N_f$ curves for the ST, 550-6, 600-6, and 650-6 samples.

the 600-6 condition attains the highest fatigue lives and fatigue strength. This apparent difference between low and high stress behavior is rationalized in Sections 4.2 and 4.3 in terms of the competition between intragranular precipitation strengthening and grain boundary weakening.

3.4. Fracture characteristics

Fig. 6 displays the macroscopic fatigue fracture surfaces of the Fe-30.5Mn-8Al-1C steel tested at stress amplitudes of 180 and 315 MPa. All samples exhibit a similar three-zone fracture morphology, comprising the crack initiation, propagation, and rapid fracture regions, with cracks initiating at the surface and propagating radially inward. At a given stress amplitude, the area fraction (fa) of the crack initiation and propagation zones (marked by yellow dashed lines in Fig. 6) is larger in the aged samples than in the ST sample; it is consistent with their extended fatigue life. The corresponding fa values were quantitatively measured and are labeled in the upper-right corner of each subfigure. Specifically, the 600-6 sample showed the largest area fraction at 180 MPa, while the 550-6 sample exhibited the highest at 315 MPa, in line with their respective optimal fatigue resistance at low and high stress amplitudes.

Fig. 7 illustrates high-magnification views of the fatigue crack propagation zone in the ST, 550-6, 600-6, and 650-6 samples tested at two stress amplitudes of 180 and 315 MPa. Well-defined fatigue striations were observed in all samples, indicating transgranular crack advancement via reversible dislocation slip on specific crystallographic planes [27]. The formation of fatigue striations typically corresponds to cyclic crack tip blunting during loading and re-sharpening during unloading, and a reduced striation spacing indicates a lower crack propagation rate [28–30]. Notably, the 550-6 and 600-6 samples exhibit more pronounced and closely spaced striations, with the narrowest spacing observed in the 550-6 sample at 315 MPa and in the 600-6 sample at 180 MPa. This corresponds to their respective longer fatigue lives shown in Fig. 5.

At the final fracture stage, the rapid fracture zones of the ST, 550-6, and 600-6 samples are predominantly characterized by dimples (Fig. 8a-c,e-g), indicating a microvoid coalescence mechanism and ductile failure. In contrast, the 650-6 sample displays mixed fracture characteristics (Fig. 8d,h), featuring both dimples and cleavage faceted regions. These facets are typically associated with reduced ductility, which is primarily attributed to the extensive precipitation of intergranular κ-carbides in the 650-6 sample.

3.5. Crack propagation behavior

To elucidate the crack-path characteristics of the fatigued samples, EBSD-based band contrast (BC) and IPF superimposed maps, Kernel average misorientation (KAM) maps, and Schmid factor (SF) maps were acquired adjacent to the main cracks at a stress amplitude of 225 MPa (Fig. 9). The BC and IPF superimposed images (Fig. 9a,d,g,j) reveal that

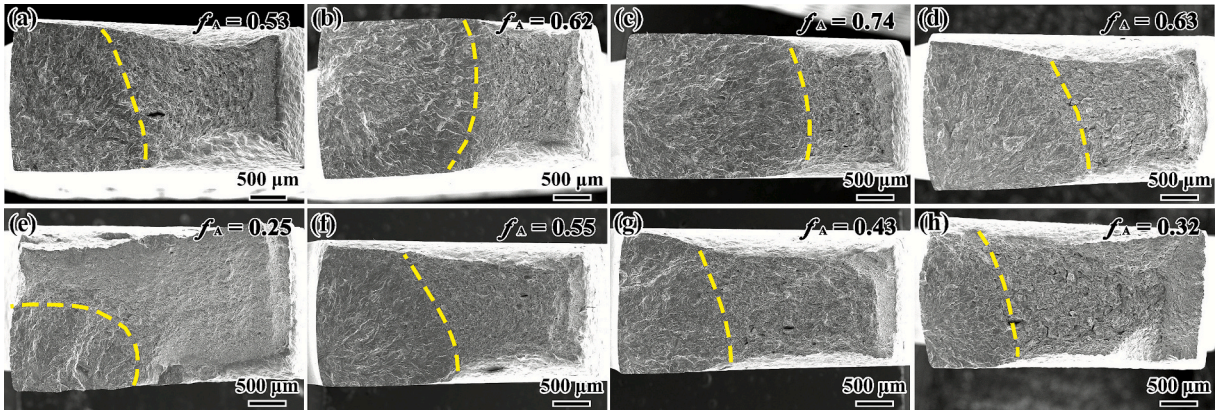

Fig. 6. SEM images showing the macroscopic fatigue fracture surfaces at different stress amplitudes of 180 MPa (a-d) and 315 MPa (e-h) for the samples of ST (a,e), 550-6 (b,f), 600-6 (c,g), and 650-6 (d,h).

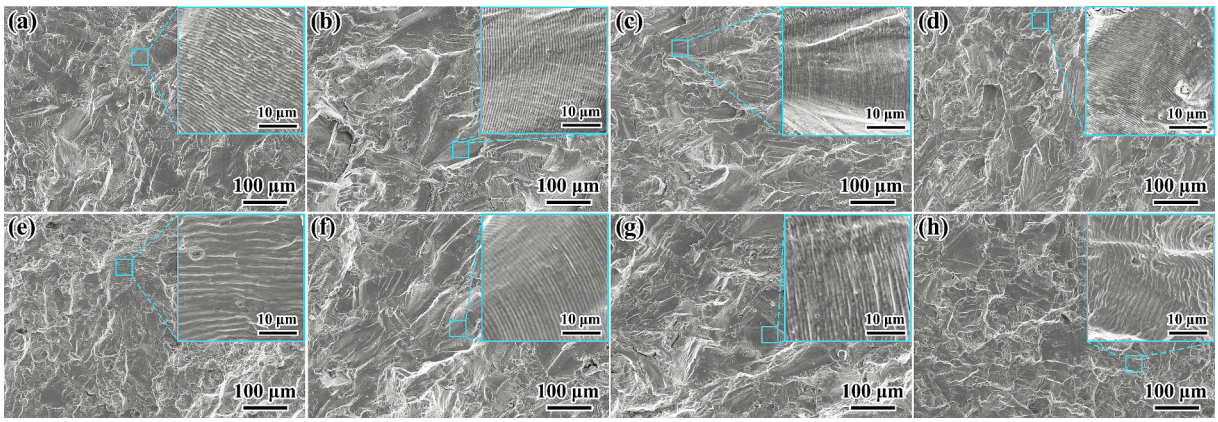

Fig. 7. SEM images showing the fracture surface morphologies in fatigue crack propagation zone at different stress amplitudes of 180 MPa (a-d) and 315 MPa (e-h) for the samples of ST (a,e), 550-6 (b,f), 600-6 (c,g), and 650-6 (d,h).

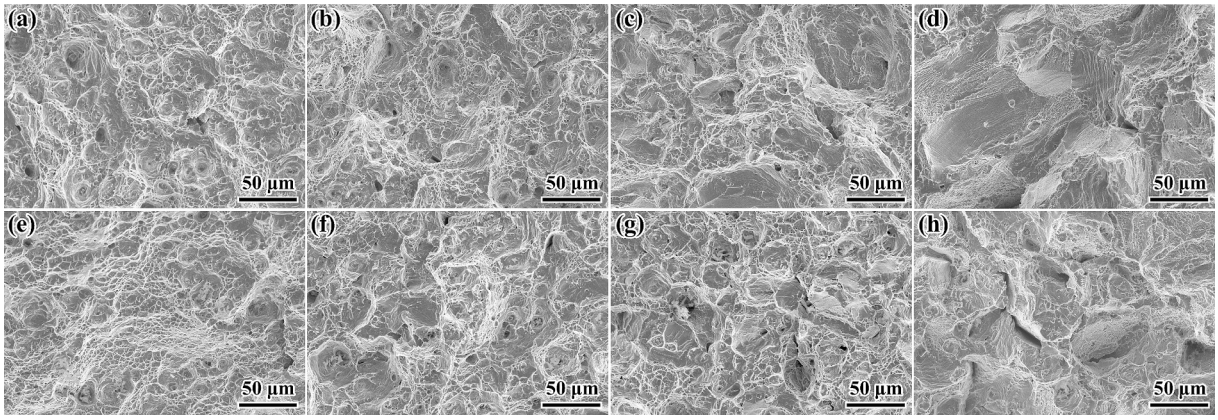

Fig. 8. SEM images showing the fracture surface morphologies in rapid fracture zone at different stress amplitudes of $\Delta\sigma/2 = 180$ MPa (a-d) and $\Delta\sigma/2 = 315$ MPa (e-h) for the samples of ST (a,e), 550-6 (b,f), 600-6 (c,g), and 650-6 (d,h).

cracks in all four samples propagate along slip bands in a transgranular mode; it is consistent with crack advance along favorably oriented {111} slip planes in face-centered cubic (FCC) alloys [31–34]. Despite this common fracture mode, clear differences in path geometry are evident. In the ST sample, the crack path within large austenite grains is relatively straight. In contrast, the aged samples (550-6, 600-6, and 650-6) exhibit significantly greater path tortuosity and frequent micro-branching as the crack interacts with the heterogeneous slip field. It should be noted, however, that the κ-carbides are nanoscale precipitates, whereas the fatigue crack path resolved by EBSD is on the micrometer scale. Given the step size and spatial resolution employed, individual intragranular κ-carbides cannot be resolved in the EBSD maps, making it impossible to establish a one-to-one correspondence between single precipitates and local crack deflections. Instead, the influence of κ-carbide precipitation is reflected at a more global level through its effect on the dominant slip mode, the degree of strain localization, and the resulting crack-path morphology. Based on the increased branching and deflection observed in the aged samples, we infer that the intragranular κ-carbides act as effective nanoscale obstacles that impede dislocation movement and redistribute the local crack-tip stress field. This process promotes the activation of multiple slip systems, thereby inducing crack deflection or branching and enhancing crack propagation resistance [35].

KAM maps (Fig. 9b,e,h,k) provide additional insight into the local plastic strain fields developed around the fatigue cracks. It is well established that the local misorientation measured by EBSD serves as a reliable quantitative indicator of micro-scale plastic strain. In the present alloy, high KAM values are primarily concentrated in the immediate vicinity of the crack path and at grain boundaries, particularly those oriented approximately perpendicular to the loading axis. This distribution reflects impeded slip transmission and the resulting dislocation pile-ups during cyclic loading. The ST sample exhibits the highest KAM intensity and the steepest gradients at these boundaries, indicative of severe strain localization. In contrast, the 550-6 and, most notably, the 600-6 samples display more moderate KAM levels with a more diffuse misorientation distribution around the crack tip. This suggests that the uniformly dispersed intragranular κ-carbides facilitate more homogeneous plastic accommodation, thereby alleviating strain incompatibility at grain boundaries. Conversely, in the 650-6 sample, localized KAM "hot spots" reappear at certain grain boundaries decorated with intergranular κ-carbides (indicated by red arrows). These hot spots point to intense local strain concentration in these regions, which is consistent with the reduced fatigue life of the 650-6 sample compared with the 600-6 and 550-6 conditions.

SF maps (Fig. 9c,f,i,l) indicate a classical sequence for crack initiation and propagation pattern, namely, cracks tend to nucleate in "hard-oriented" grains (low SF) due to limited slip activity and high local stress concentration, and then propagate into adjacent "soft-oriented" grains (high SF) where dislocation slip is more easily activated, providing a low-energy path for crack advancement. Overall, the κ-carbide–induced tortuosity/branching, combined with the reduced grain-boundary strain localization, collectively enhance the crack-propagation resistance in the aged samples. A comparable microstructural scenario has been reported in Ni-based IN718, where nanoscale coherent γ" precipitates modify the slip mode and crack-path tortuosity during fatigue crack growth, thereby strongly affecting the fatigue crack propagation

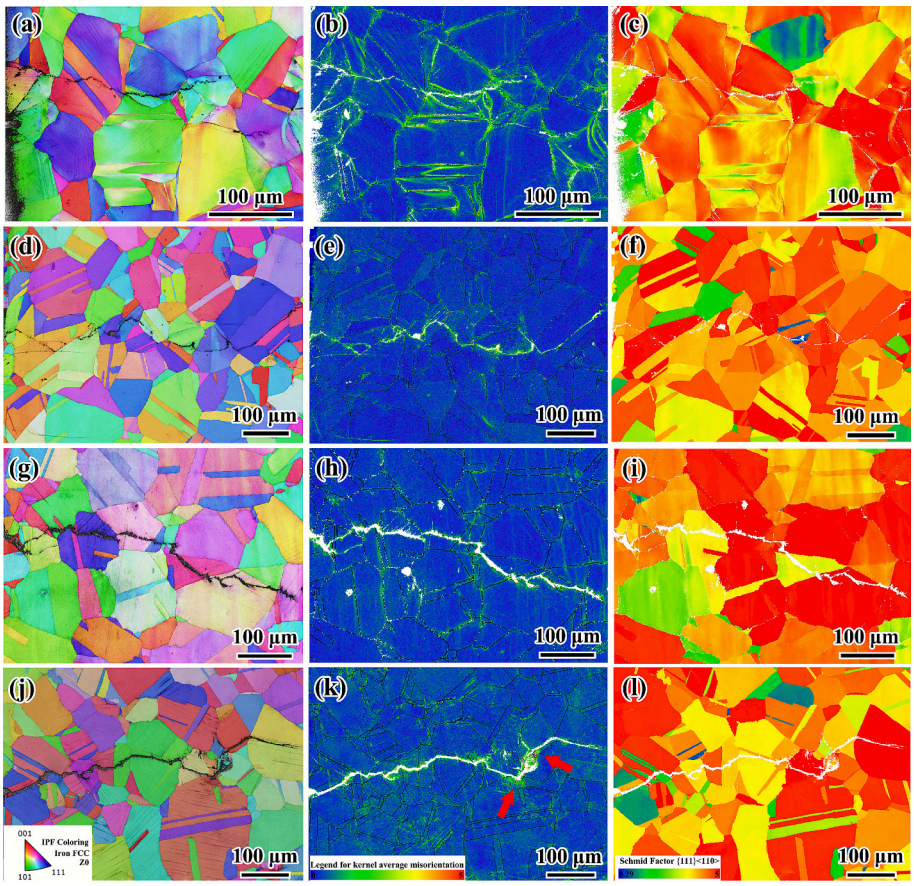

Fig. 9. EBSD analysis of fatigue crack propagation paths for ST (a-c), 550-6 (d-f), 600-6 (g-i), and 650-6 (j-l) samples tested at a stress amplitude of 225 MPa. (a,d,g,j) BC and IPF superimposed maps, (b,e,h,k) KAM maps, and (c,f,i,l) SF maps.

resistance [36].

3.6. Surface deformation characteristics

Representative SEM images of the surface deformation features in the ST and 600-6 samples fatigued at stress amplitudes of 180 MPa and

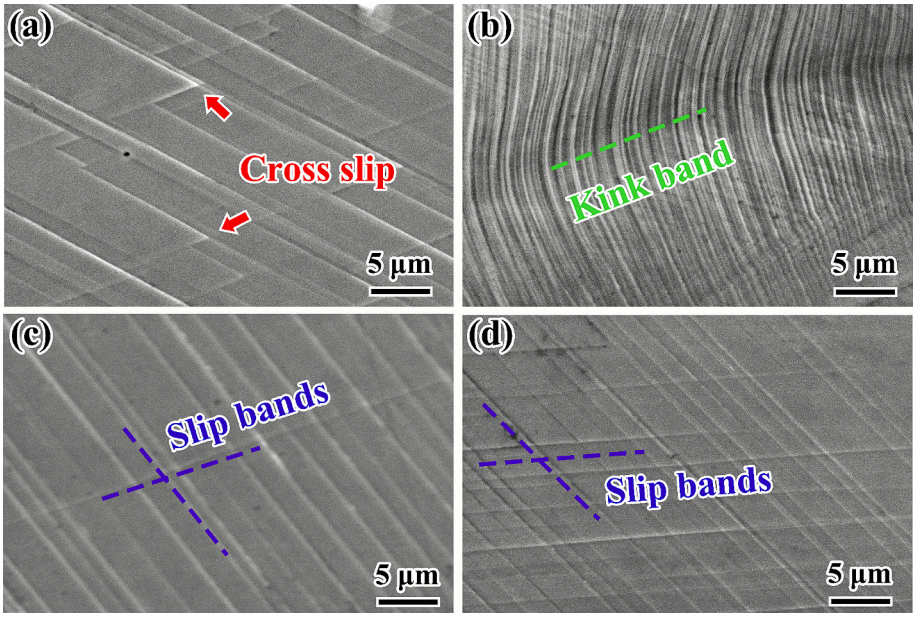

Fig. 10. SEM micrographs showing the surface deformation morphologies in the ST (a,b) and 600-6 (c,d) samples fatigued at $\Delta\sigma/2 = 180$ MPa (a,c) and $\Delta\sigma/2 = 315$ MPa (b,d).

315 MPa are shown in Fig. 10. In the ST sample, local wavy slip (Fig. 10a) is observed at $\Delta\sigma/2 = 180$ MPa. The wavy slip morphology results from screw dislocations undergoing cross slip after pile-up, leading to the formation of step-like slip bands. This irreversible slip intensifies local strain concentration and accelerates crack initiation [37,38]. At the higher stress amplitude of 315 MPa, stress localization in the ST sample becomes more severe and kink bands are formed (Fig. 10b), indicative of a loss of deformability due to highly localized and irreversible deformation.

In contrast, the 600-6 sample exhibits distinct, uniformly spaced slip bands at both stress amplitudes of 180 MPa and 315 MPa (Fig. 10c,d). These slip bands are actually planar slip bands that are attributed to the "glide plane softening" mechanism as discussed in Section 4.1. At $\Delta\sigma/2 = 315$ MPa, the spacing between slip bands decreases and the cooperative activation of multiple slip systems accommodates plastic strain more uniformly. As a result, deformation proceeds via a dense yet largely reversible network of slip bands, effectively preventing catastrophic strain localization such as kink bands [19,38,39]. In summary, the presence of intragranular κ-carbides promotes reversible planar slip and thereby enhances the fatigue deformation response of the 600-6 sample across the studied cyclic stress levels.

3.7. Deformation microstructures

Fig. 11 shows the typical dislocation structures in the ST and aged samples (550-6, 600-6, 650-6) fatigued at $\Delta\sigma/2$ of 180 and 315 MPa. Planar dislocation slip was observed to be the dominant deformation mechanism in all cases. At $\Delta\sigma/2 = 180$ MPa, the ST sample exhibits well-developed dislocation bands composed of dense dislocation arrays aligned along primary slip planes (Fig. 11a), namely persistent Lüder's bands (PLBs), which are indicative of strain localization in fatigued samples. In contrast, the aged samples exhibit a substantially lower overall dislocation density and more uniformly distributed planar slip bands (Fig. 11c,e,g). This difference arises from the fine intragranular κ-carbides, which dynamically interact with gliding dislocations during cycling, impede their slip and accumulation, and thereby suppress the

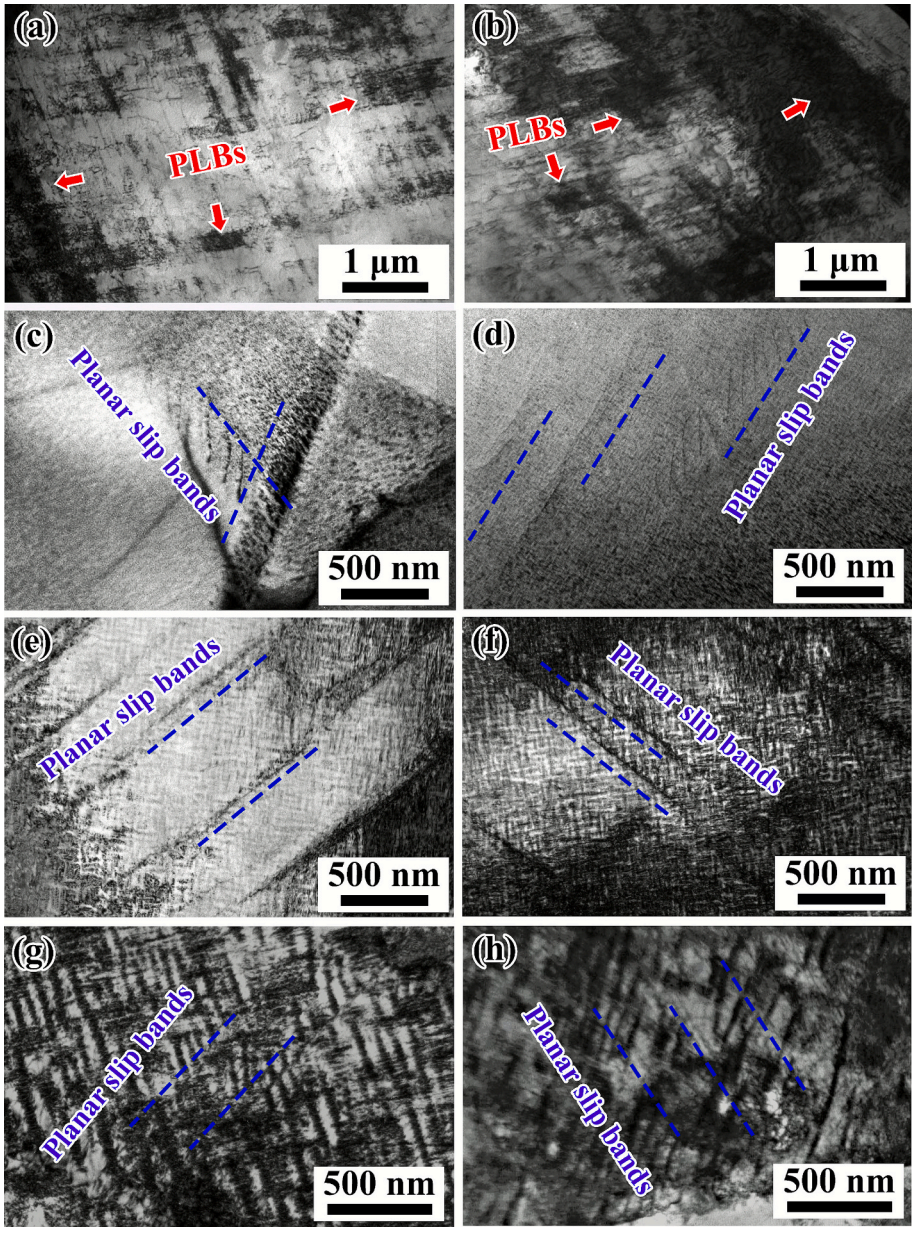

Fig. 11. Bright-field TEM micrographs showing the deformation microstructures in the ST (a,b), 550-6 (c,d), 600-6 (e,f), and 650-6 (g,h) samples fatigued at $\Delta\sigma/2 = 180$ MPa (a,c,e,g) and $\Delta\sigma/2 = 315$ MPa (b,d,f,h).

formation and coalescence of localized bands. At the higher stress amplitude of $\Delta\sigma/2 = 315$ MPa (Fig. 11b,d,f,h), all samples exhibit similar features to those at $\Delta\sigma/2 = 180$ MPa, but with narrower slip band spacing and more extensive dislocation movement. The ST sample shows more intensely localized bands, whereas the aged samples maintain a comparatively uniform network of planar bands. These microstructural observations align with the improved fatigue resistance of the aged samples, particularly 600-6, by limiting strain localization and delaying damage evolution.

4. Discussion

4.1. Impact of intragranular κ-carbides on dislocation slip mode

In the ST sample, the observed planar slip bands are primarily attributed to the short-range ordering (SRO) structure within the austenitic matrix. This observation aligns with the "glide plane softening" effect proposed by Gerold and Karnthaler [40]. The strong C-Mn chemical affinity leads to a non-random solute distribution, forming locally ordered clusters [41–43]. When a leading dislocation cuts through such an SRO region, it disrupts the local order and reduces the resistance on that specific glide plane, thereby facilitating subsequent coplanar glide and the formation of distinct planar slip bands [44–46]. However, because SRO structure is relatively weak and spatially heterogeneous, repeated shearing by moving dislocations tends to destroy it prematurely. The resulting non-uniform softening across potential slip planes fosters local strain concentrations (Fig. 12a).

Aging treatment induces coherent ordered $L1_2$-structured intragranular κ-carbides that amplify "glide plane softening" effect beyond the SRO level. When moving dislocations shear these precipitates [47], anti-phase boundaries (APBs) are created. After a plane has been swept and order locally disrupted, subsequent dislocations traverse the same plane with reduced resistance because additional APBs are no longer formed [42]. This path-dependent process channels dislocation activity to a limited set of previously softened glide planes, strengthening slip planarity and suppressing cross-slip. The uniform nanoscale dispersion of κ-carbides promotes the homogeneous formation of planar slip bands and distributes plastic strain over multiple coplanar systems, thereby mitigating localized strain concentrations. Crucially, this enhanced slip planarity and restricted cross-slip significantly improve the reversibility of dislocation motion under cyclic loading [44]. Dislocations can more readily retrace their path on well-defined softened planes during load reversal, leading to reduced accumulated damage per cycle (Fig. 12b). These mechanisms rationalize the superior fatigue performance of the aged samples relative to the ST samples.

4.2. Effect of intragranular κ-carbides on fatigue behavior at low stress amplitudes

At low stress amplitudes, the fatigue crack initiation stage occupies a significant portion of the total fatigue life; strengthening the matrix is therefore an effective route to delay initiation [48,49]. In Fe-Mn-Al-C alloys, precipitation strengthening by intragranular κ-carbides is dominant, and the active dislocation-precipitate interaction depends on the size of precipitates. The dominant mechanism transitions from shearing to Orowan bypassing as the precipitate size increases, and a critical size of approximately 27 nm was reported in analogous alloy systems [50]. In the present work, intragranular κ-carbides in both the 550-6 and 650-6 samples (with average sizes of ~11 nm and ~24 nm, respectively) remain in the shearing regime (Fig. 13a,b). Fig. 13(c,d) present high-resolution TEM (HRTEM) images of κ-carbides in the fatigued 550-6 and 650-6 samples, respectively. Fig. 13(c',d') are magnified views of the areas marked by green rectangles in Fig. 13(c,d) and the corresponding inverse fast Fourier transform (IFFT) filtered images of the selected regions are shown in Fig. 13(c",d"), where the κ-carbides, austenitic matrix, and transition regions are delineated in white, black, and blue, respectively. The presence of shear steps and localized contrast variations within the carbides (indicated by red arrows) suggests a disruption of the ordered structure and the formation of APBs, which is consistent with the view that repeated dislocation shearing causes mechanical dissolution of κ-carbides during cyclic loading, as also reported by Wegener et al. [22].

Under shearing, the strengthening increment includes the contributions from chemical strengthening ($\Delta\sigma_{chs}$), stacking fault strengthening ($\Delta\sigma_{sfs}$), coherency strain strengthening ($\Delta\sigma_{cs}$), modulus mismatch strengthening ($\Delta\sigma_{ms}$), and order strengthening ($\Delta\sigma_{os}$). Given the negligible chemical effect and the absence of observable stacking faults here, $\Delta\sigma_{chs}$ and $\Delta\sigma_{sfs}$ are minor. The remaining increments are summarized as below [50]:

$$\Delta\sigma_{cs} = KM(G\varepsilon_c)^{\frac{3}{2}}\left(\frac{rf}{0.5Gb}\right)^{\frac{1}{2}} \tag{2}$$

$$\Delta\sigma_{ms} = 0.0055M(\Delta G)^{\frac{3}{2}}\left(\frac{2f}{G}\right)^{\frac{1}{2}}\left(\frac{r}{b}\right)^{\frac{3m}{2}-1} \tag{3}$$

$$\Delta\sigma_{os} = 0.81M\frac{\gamma_{APB}}{2b}\left(\frac{3\pi f}{8}\right)^{\frac{1}{2}} \tag{4}$$

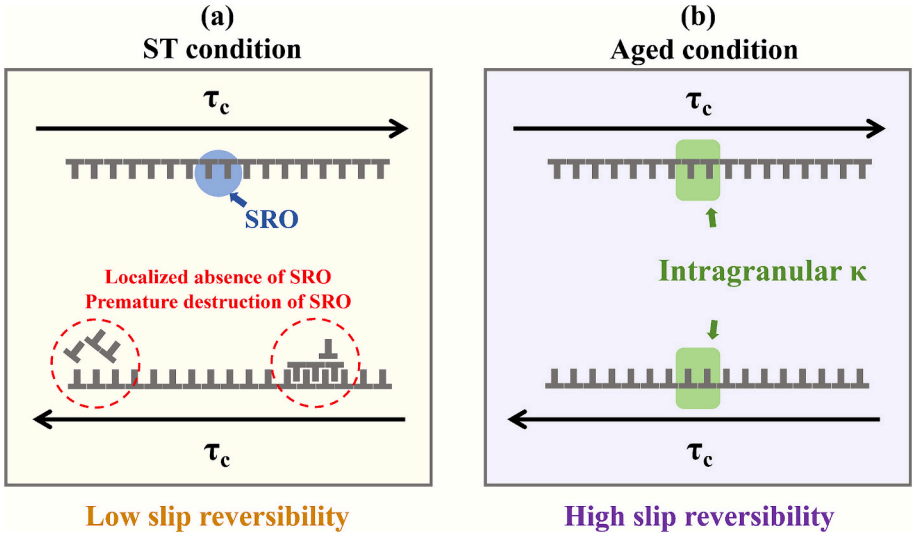

Fig. 12. Schematic diagram showing the difference in slip reversibility in the ST (a) and aged (b) samples.

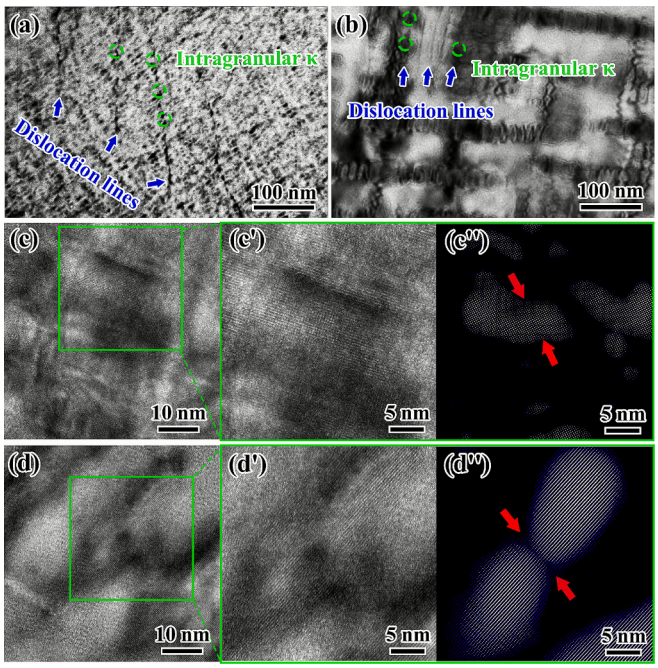

Fig. 13. TEM analysis of dislocation-carbide interactions in the samples fatigued at $\Delta\sigma/2 = 180$ MPa. (a,b) Bright-field TEM images of κ-carbides in the fatigued 550-6 (a) and 650-6 (b) samples; (c,d) HRTEM of κ-carbides and dislocations in the fatigued 550-6 (c) and 650-6 (d) samples; (c') and (d') are magnified views of the areas marked by green rectangles in (c) and (d), respectively; (c") and (d") are the corresponding IFFT filtered images of the areas in (c') and (d'), respectively, where the κ-carbides, austenitic matrix, and transition regions are highlighted in white, black, and blue, respectively.

where K and m are constants equal to 2.6 and 0.85, respectively, M is the Taylor factor, ε_c is the lattice misfit strain, b is the Burgers vector, G is the shear modulus, ΔG is the modulus misfit between γ-matrix and κ-carbide, γ_{APB} is the anti-phase boundary energy, and r and f correspond to the radius and volume fraction of intragranular κ-carbides, respectively [50]. Notably, $\Delta\sigma_{cs}$ and $\Delta\sigma_{ms}$ act prior to shearing and increase with precipitate size at a fixed f.

Based on the above analyses, non-monotonic fatigue lives (or fatigue strength) ranking in different samples at low stress amplitudes can be well interpreted. The ST sample, lacking intragranular κ-carbides, exhibits the lowest fatigue strength. Among the aged samples, the 600-6 condition, with an intermediate intragranular κ-carbide size of ~18 nm, achieves the highest fatigue strength due to a favorable balance of shearing-related strengthening and uniform planar slip. The 550-6 sample, containing the finest intragranular κ-carbides (~11 nm), ranks the third because smaller precipitates provide less $\Delta\sigma_{cs}$ and $\Delta\sigma_{ms}$. Although the 650-6 sample possesses the largest intragranular κ-carbides (~24 nm), its fatigue strength is only the second-best. This deviation from a monotonic size-strengthening trend is attributed to severe intergranular precipitation prevalent in the 650-6 condition, which undermines the beneficial intragranular effect; it will be further discussed in the following Section 4.3.

4.3. Effect of intergranular κ-carbides on fatigue behavior at high stress amplitudes

At high stress amplitudes, the activation of mobile dislocations becomes more pronounced, and their interactions with grain boundaries are more significant. Thus, the state of grain boundaries exerts a strong influence on fatigue life [51]. Plastic deformation heterogeneity in metallic materials predominantly arises from SFs mismatch between adjacent grains. "Soft-oriented" grains (high SF) can activate dislocation slip at lower applied stress compared with "hard-oriented" grains (low SF). According to Schmid's law, the critical resolved shear stress (τ_c) required for slip system activation in FCC metals satisfies:

$$\sigma_a = \tau_c / \mu \tag{5}$$

where μ is the Schmid factor and the σ_a is the applied tensile stress. During cyclic deformation, dislocations are preferentially multiplicated and glide in soft-oriented grains, leading to pile-ups against boundaries shared with hard-oriented neighbors.

The dislocation distribution is clearly demonstrated by TEM characterizations of microstructures adjacent to grain boundaries after fatigue at a $\Delta\sigma/2$ of 315 MPa (Fig. 14). In the 550-6 sample (Fig. 14a), κ-carbides are uniformly distributed within the grains, with no coarse precipitation at the grain boundaries. These intragranular κ-carbides facilitate organized planar slip bands, promoting relatively uniform slip transfer across grain boundaries [26]. By contrast, the 650-6 sample exhibits coarse intergranular κ-carbide precipitation disrupting the continuity of grain boundaries (Fig. 14b). These brittle particles act as strong slip barriers, intensifying dislocation pile-ups, promoting the formation of tangled dislocation structures with wavy-slip patterns, elevating local stress concentration, and eventually leading to earlier crack nucleation and shortened fatigue life.

Based on the above analyses, the influence of intergranular κ-carbides on the fatigue deformation mechanism is schematically illustrated in Fig. 15. These differences in slip behavior directly correlate with the fatigue performance at high stress amplitudes. In the 550-6 sample, which has a homogeneous intragranular κ-carbide distribution and an absence of intergranular κ-carbides, dislocation slip transfer is facilitated and localization is suppressed. In the 650-6 sample, the presence of severe intergranular κ-carbides impedes dislocation slip at grain boundaries to cause severe dislocation pile-ups, exacerbates local stress concentrations, and accelerates strain localization, leading to premature crack initiation and failure. Consistently, the fatigue life ranking at high amplitudes is 550-6 sample of best, 600-6 of intermediate, and 650-6 of poorest, tracking the increasing severity of intergranular precipitation. Apparently, such a detrimental grain boundary effect that offsets the size benefit at low stress amplitudes discussed in Section 4.2 also degrades fatigue resistance at high stress amplitudes by the same mechanism.

Based on the above analyses, achieving an optimal κ-carbide precipitation state is essential for enhancing the fatigue performance of Fe-Mn-Al-C austenitic low-density steels. This requires a critical balance, namely, promoting a uniform distribution of intragranular κ-carbides with an appropriate size, while effectively suppressing detrimental precipitation at grain boundaries. Beneficial intragranular precipitates improve fatigue strength and fatigue life at low stress amplitudes by enhancing precipitation strengthening and dislocation slip reversibility. In contrast, coarse intergranular κ-carbides significantly degrade fatigue life at high stress amplitudes by accelerating stress concentration. Thus, the superior overall fatigue properties can be effectively achieved through microstructural control that maximizes the advantages of intragranular precipitation and simultaneously minimizes the harmful effects of intergranular precipitation.

5. Conclusions

This study systematically investigated the tension–tension fatigue behavior of an Fe-30.5Mn-8Al-1C low-density steel in both ST and aged conditions, focusing on the critical role of κ-carbides in governing fatigue performance, and the main conclusions can be drawn as follows.

(1) In both ST and aged conditions, planar dislocation slip occurs during cyclic deformation due to the "glide plane softening" effect. In ST, SRO-driven softening is weak and spatially heterogeneous, promoting cross-slip, to a certain extent, and local strain concentration. By contrast, uniformly dispersed intragranular

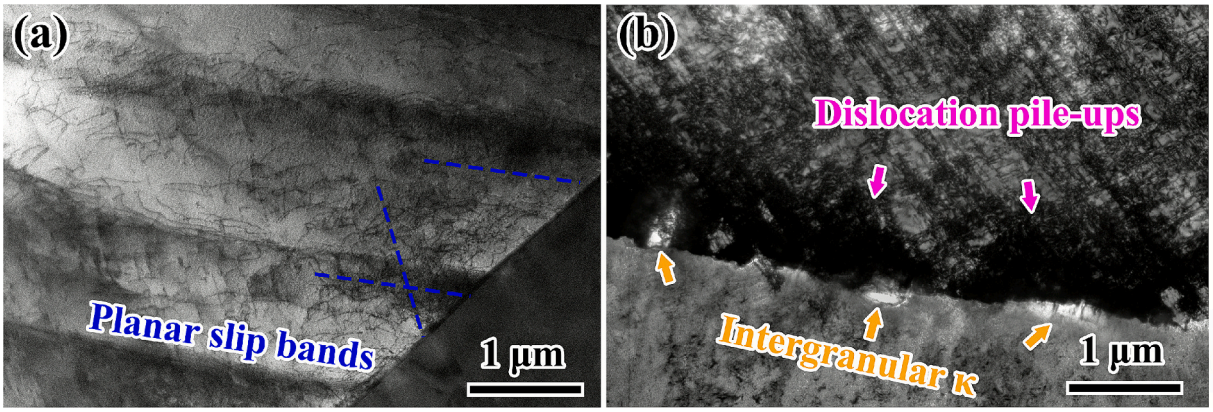

Fig. 14. Bright-field TEM micrographs showing the deformation microstructures near grain boundaries in the 550-6 (a) and 650-6 (b) samples fatigued at $\Delta\sigma/2 = 315$ MPa.

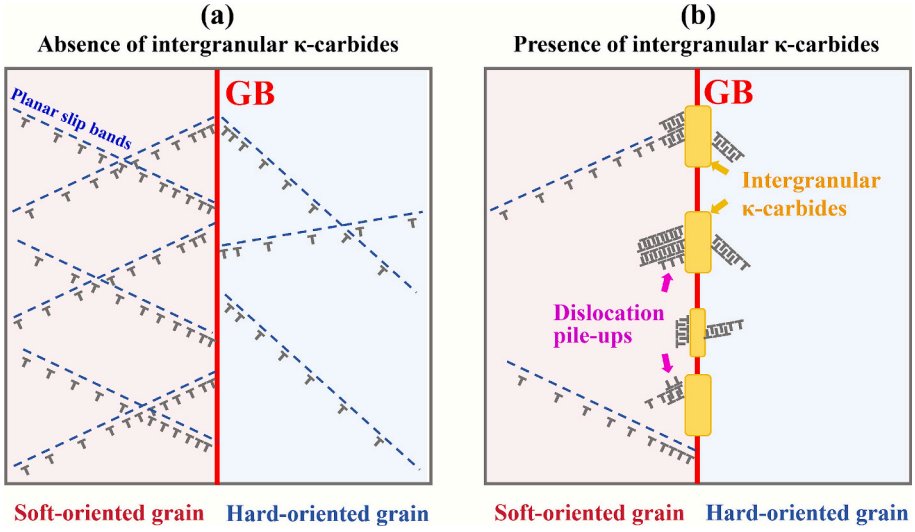

Fig. 15. Schematic diagram showing the impact of absence (a) and presence (b) of intergranular κ-carbides on the cyclic deformation behavior of Fe-Mn-Al-C alloys.

κ-carbides enhance the "glide plane softening" effect through repeated shearing by moving dislocations, suppress cross-slip, increase the reversibility of dislocation motion, and consequently improve fatigue life.

(2) Under the present conditions, the precipitation strengthening contributing from κ-carbides is primarily governed by a shearing mechanism, and the strengthening effect becomes enhanced with increasing precipitate size. Therefore, an appropriate increase in the size of κ-carbides can effectively improve fatigue strength and fatigue life at low stress amplitudes.

(3) At high stress amplitudes, the precipitation of intergranular κ-carbides markedly deteriorates the fatigue life of Fe-Mn-Al-C alloys. These intergranular κ-carbides impede slip transfer, promote dislocation pile-ups and localized stress concentration, thereby accelerating fatigue damage accumulation. Consequently, a key strategy for enhancing fatigue performance at high stress amplitudes of Fe-Mn-Al-C alloys is to promote intragranular precipitation while suppressing its formation at grain boundaries.

(4) Among the aging conditions investigated, the 600-6 sample (aged at 600 °C for 6 h) exhibits the best overall mechanical performance, combining high tensile strength with superior fatigue strength and fatigue life.

CRediT authorship contribution statement

J.H. Du: Writing – original draft, Investigation. **P. Chen:** Writing – original draft, Investigation, Funding acquisition, Conceptualization. **Z. P. Jia:** Investigation. **X.W. Li:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw/processed data required to reproduce these findings cannot

be shared at this time as the data also forms part of an ongoing study.

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